Why do transfers fail?

Luck, usage and chemistry in Big-5 football, per 90 minutes, 2014–2025

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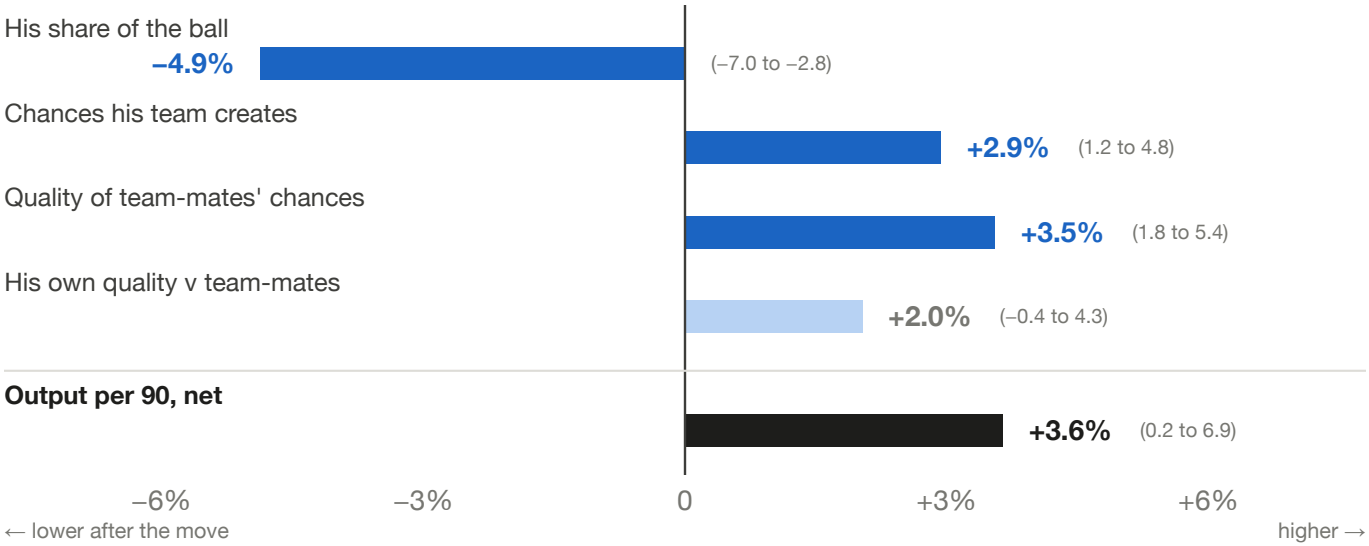
Moving club costs an attacker about 5% of his share of the ball, and the better team he joins makes up for it: per 90 he ends up 3.6% ahead of players who stayed. Losing the ball is about a quarter of a flop (output per 90 down by more than 25%). Movers flop 19% of the time against 18% for stayers, and a luck-only benchmark gives 22%. Midfielders pay about twice as much of the ball and end up behind.

–4.9%
share of the ball after a move

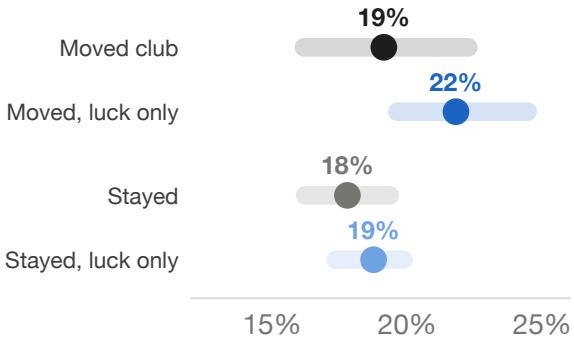
–1.7%
chance quality, 5 points more ball

+0.5%
pair chemistry, second season v first

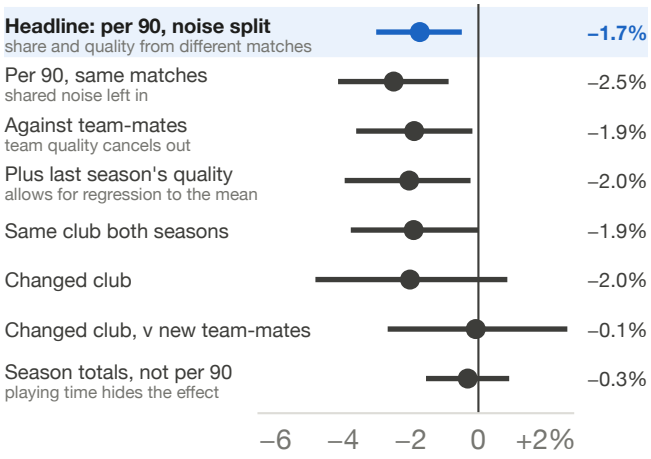
What a typical move does, against players who stayed (per 90)



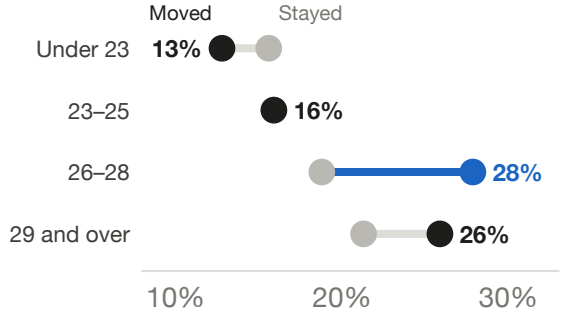
Flop rates against luck alone



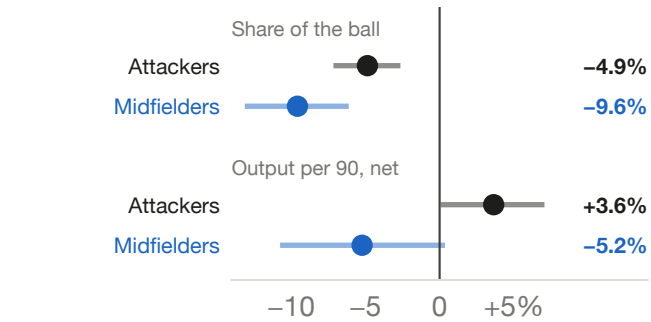
Same player, 5 points more of the ball



Flop rate by age: moved v stayed



After a move: attackers v midfielders



Limits: shots and shot-assists only, so it suits attackers; regulars only (900+ minutes in both seasons); observational, so it shows patterns, not causes. Data: Understat xG; Transfermarkt line-ups and substitutions. Intervals are 95% and resample whole players.

github.com/benedictprimmer-web/football-usage-efficiency